

1. Analytical Derivation of Sphere-of-Influence Encounter Points

1.1. Problem Statement

Given two Keplerian elliptical orbits (spacecraft orbit 1, planet orbit 2) in heliocentric ecliptic coordinates, find the eccentric anomalies E_1, E_2 such that the distance between the spacecraft and the planet equals the planet's sphere-of-influence radius r_{SOI} :

$$\|A\mathbf{r}_1(E_1) - B\mathbf{r}_2(E_2)\| = r_{\text{SOI}} \quad (1)$$

where A and B are known constant 3×2 rotation matrices mapping from orbital-plane 2D coordinates to 3D heliocentric ecliptic coordinates.

1.2. Notation

- a_i, b_i, e_i : semi-major axis, semi-minor axis, and eccentricity of orbit i (with $b_i = a_i \sqrt{1 - e_i^2}$)
- E_i : eccentric anomaly of orbit i
- $A \in \mathbb{R}^{3 \times 2}, B \in \mathbb{R}^{3 \times 2}$: orbital-to-ecliptic rotation matrices with components A_{jk}, B_{jk}
- r_{SOI} : radius of the planet's sphere of influence

1.3. Position in Orbital Plane

For a Keplerian ellipse, the 2D position vector in the orbital plane as a function of eccentric anomaly is:

$$\mathbf{r}_i(E_i) = \begin{pmatrix} a_i(\cos E_i - e_i) \\ b_i \sin E_i \end{pmatrix} \quad (2)$$

1.4. Distance Constraint

Square both sides of the constraint:

$$(A\mathbf{r}_1 - B\mathbf{r}_2)^\top (A\mathbf{r}_1 - B\mathbf{r}_2) = r_{\text{SOI}}^2 \quad (3)$$

Expand the left-hand side:

$$\mathbf{r}_1^\top A^\top A \mathbf{r}_1 - 2\mathbf{r}_1^\top A^\top B \mathbf{r}_2 + \mathbf{r}_2^\top B^\top B \mathbf{r}_2 = r_{\text{SOI}}^2 \quad (4)$$

1.4.1. Orthonormality of A and B

Since A and B are 3×2 matrices whose columns are orthonormal (they embed an orthonormal frame from the orbital plane into 3D space):

$$A^\top A = I_{2 \times 2}, \quad B^\top B = I_{2 \times 2} \quad (5)$$

Therefore the self-terms simplify:

$$\mathbf{r}_1^\top A^\top A \mathbf{r}_1 = \mathbf{r}_1^\top \mathbf{r}_1 = |\mathbf{r}_1|^2 =: r_1^2 \quad (6)$$

$$\mathbf{r}_2^\top B^\top B \mathbf{r}_2 = \mathbf{r}_2^\top \mathbf{r}_2 = |\mathbf{r}_2|^2 =: r_2^2 \quad (7)$$

1.4.2. The coupling matrix C

Define the 2×2 matrix:

$$C := 2A^\top B \quad (8)$$

This encodes the entire mutual orientation of the two orbital planes. Equation 4 becomes:

$$r_1^2 + r_2^2 - \mathbf{r}_1^\top C \mathbf{r}_2 = r_{\text{SOI}}^2 \quad (9)$$

Note: C is **not** necessarily symmetric — it is a general 2×2 real matrix with $|C_{jk}| \leq 2$.

1.5. Expanding the Scalar Terms

1.5.1. Heliocentric distances r_i^2

$$r_i^2 = a_i^2(1 - e_i \cos E_i)^2 \quad (10)$$

1.5.2. Cross-term $\mathbf{r}_1^\top \mathbf{C} \mathbf{r}_2$

Decompose $\mathbf{r}_i$ into its centered-ellipse part and eccentricity offset:

$$\mathbf{r}_i(E_i) = \mathbf{r}_{0i}(E_i) - \mathbf{a}_i, \quad \mathbf{r}_{0i} = \begin{pmatrix} a_i \cos E_i \\ b_i \sin E_i \end{pmatrix}, \quad \mathbf{a}_i = \begin{pmatrix} a_i e_i \\ 0 \end{pmatrix} \quad (11)$$

$$\mathbf{r}_1^\top \mathbf{C} \mathbf{r}_2 = (\mathbf{r}_{01} - \mathbf{a}_1)^\top \mathbf{C} (\mathbf{r}_{02} - \mathbf{a}_2) = \mathbf{r}_{01}^\top \mathbf{C} \mathbf{r}_{02} - \mathbf{a}_1^\top \mathbf{C} \mathbf{r}_{02} - \mathbf{r}_{01}^\top \mathbf{C} \mathbf{a}_2 + \mathbf{a}_1^\top \mathbf{C} \mathbf{a}_2 \quad (12)$$

1.5.3. Constant sub-expressions

Since $\mathbf{C}$, $\mathbf{a}_1$, and $\mathbf{a}_2$ are all constants (independent of E_1, E_2), we introduce the following constants:

$$\mathbf{p}^\top := \mathbf{a}_1^\top \mathbf{C} \in \mathbb{R}^2, \quad \mathbf{q} := \mathbf{C} \mathbf{a}_2 \in \mathbb{R}^2, \quad s := \mathbf{a}_1^\top \mathbf{C} \mathbf{a}_2 \quad (13)$$

$$\mathbf{r}_1^\top \mathbf{C} \mathbf{r}_2 = \mathbf{r}_{01}^\top \mathbf{C} \mathbf{r}_{02} - \mathbf{p}^\top \mathbf{r}_{02} - \mathbf{r}_{01}^\top \mathbf{q} + s \quad (14)$$

$$r_1^2 + r_2^2 - \mathbf{r}_1^\top \mathbf{C} \mathbf{r}_2 = a_1^2(1 - e_1 \cos E_1)^2 + a_2^2(1 - e_2 \cos E_2)^2 - \mathbf{r}_{01}^\top \mathbf{C} \mathbf{r}_{02} + \mathbf{p}^\top \mathbf{r}_{02} + \mathbf{r}_{01}^\top \mathbf{q} - s \quad (15)$$

1.6. Stationarity Conditions

To find the closest approach, minimize the squared distance:

$$f(E_1, E_2) := r_1^2 + r_2^2 - \mathbf{r}_1^\top \mathbf{C} \mathbf{r}_2 \quad (16)$$

The eccentric-anomaly derivatives of the centered position vectors are:

$$\dot{\mathbf{r}}_{0i} := \frac{\partial \mathbf{r}_{0i}}{\partial E_i} = \begin{pmatrix} -a_i \sin E_i \\ b_i \cos E_i \end{pmatrix} \quad (17)$$

Since $\mathbf{a}_i$ is constant, $\frac{\partial \mathbf{r}_i}{\partial E_i} = \dot{\mathbf{r}}_{0i}$.

1.6.1. Partial derivatives

Setting the partial derivatives to zero and using $\frac{\partial r_i^2}{\partial E_i} = 2a_i^2 e_i \sin E_i (1 - e_i \cos E_i)$:

$$\frac{\partial f}{\partial E_1} = 2a_1^2 e_1 \sin E_1 (1 - e_1 \cos E_1) - \dot{\mathbf{r}}_{01}^\top \mathbf{C} \mathbf{r}_2 = 0 \quad (18)$$

$$\frac{\partial f}{\partial E_2} = 2a_2^2 e_2 \sin E_2 (1 - e_2 \cos E_2) - \mathbf{r}_1^\top \mathbf{C} \dot{\mathbf{r}}_{02} = 0 \quad (19)$$

1.6.2. Scaled coupling constants

To expand the bilinear terms into trigonometric form, define the four scaled coupling constants:

$$\alpha := a_1 a_2 C_{11}, \quad \beta := a_1 b_2 C_{12}, \quad \gamma := b_1 a_2 C_{21}, \quad \delta := b_1 b_2 C_{22} \quad (20)$$

The cross-terms in Equation 18 and Equation 19 expand as:

$$\dot{\mathbf{r}}_{01}^\top \mathbf{C} \mathbf{r}_2 = (-\alpha \sin E_1 + \gamma \cos E_1)(\cos E_2 - e_2) + (-\beta \sin E_1 + \delta \cos E_1) \sin E_2 \quad (21)$$

$$\mathbf{r}_1^\top \mathbf{C} \dot{\mathbf{r}}_{02} = (-\alpha \sin E_2 + \beta \cos E_2)(\cos E_1 - e_1) + (-\gamma \sin E_2 + \delta \cos E_2) \sin E_1 \quad (22)$$

The full stationarity conditions are therefore:

$$2a_1^2 e_1 \sin E_1 (1 - e_1 \cos E_1) = (-\alpha \sin E_1 + \gamma \cos E_1)(\cos E_2 - e_2) + (-\beta \sin E_1 + \delta \cos E_1) \sin E_2 \quad (23)$$

$$2a_2^2 e_2 \sin E_2 (1 - e_2 \cos E_2) = (-\alpha \sin E_2 + \beta \cos E_2)(\cos E_1 - e_1) + (-\gamma \sin E_2 + \delta \cos E_2) \sin E_1 \quad (24)$$